GRADE 7 · CAMBRIDGE SCIENCE · EXAM REVIEW

The Science Study Guide

Everything on your review worksheet, explained in plain English, with pictures, memory tricks and the answers to every question.

UNIT 4

Grouping & Identifying Organisms

What makes something alive, why viruses are weird, species, and how to use a key.

UNIT 5

Properties of Materials

Metals and non-metals, alloys, acids and alkalis, the pH scale and hazard symbols.

UNIT 9

Electricity

Current, circuit symbols, ammeters, series circuits, conductors and insulators.

How to use this guide

Read one unit at a time. Each unit starts with the **big idea**, then explains every topic from your worksheet. At the end of each unit there is a table with **all the worksheet answers** so you can check yourself. The very last page is a **one-page cheat sheet** for the night before the exam.

EXAM TIP
How to get the marks.

WATCH OUT!
A common mistake to avoid.

MEMORY TRICK
An easy way to remember.

REAL LIFE
Where you see this every day.

4 Grouping & Identifying Organisms

Living or not living? Same species or not? And how do scientists sort millions of living things?

Big idea: All living things do the same **seven life processes**. Scientists use these to decide what is alive, and they sort living things into **groups** (like species) so the huge variety of life is easier to study.

4.1 The seven life processes (MRS GREN)

To count as **alive**, an organism must be able to do **all seven** of these things. An **organism** is just a fancy word for "a living thing".

MEMORY TRICK

Meet **MRS GREN**. Each letter is one life process: **M**ovement, **R**espiration, **S**ensitivity, **G**rowth, **R**eproduction, **E**xcretion, **N**utrition.

	Life process	What it means	Example
M	**Movement**	Changing the shape or position of part or all of the body.	A cheetah running; a sunflower turning to face the Sun.
R	**Respiration**	Breaking down nutrients (food) inside cells to **release energy**.	Your cells break down sugar from breakfast to give you energy for football.
S	**Sensitivity**	Detecting changes in the environment and responding to them.	A polar bear **smells** meat from far away with its nose.
G	**Growth**	A **permanent** increase in size and mass. Young plants and animals get bigger.	A baby zebra growing into an adult zebra.
R	**Reproduction**	Making new individuals of the **same kind** of organism ("producing young ones").	A zebra having a foal; a plant making seeds.
E	**Excretion**	Getting rid of **waste materials the body has made**.	Animals excrete **carbon dioxide** when they breathe out.
N	**Nutrition**	Feeding: taking in materials for energy and growth. **Another word for nutrition is "feeding".**	A lion eating meat; a plant making its own food.

WATCH OUT!

Respiration is NOT the same as breathing. Breathing just moves air in and out. Respiration is the chemical process in every cell that releases energy from food. A plant respires but has no lungs!

WATCH OUT!
"Growth" must be permanent. A balloon getting bigger, or you getting puffy after a big lunch, is not growth. Growth is a lasting increase in size **and** mass.

4.2 Plants vs animals

How plants get food

Plants make their own food by **photosynthesis**. They need three things:

- ☀️ **Sunlight** (energy)
- 💧 **Water** (from the soil, through the roots)
- 💨 **Carbon dioxide** (from the air, through the leaves)

So an **oak tree** can make its own food, but a mushroom, an earthworm and a lion cannot.

WATCH OUT!
A **mushroom is NOT a plant**. It is a fungus and cannot use sunlight to make food.

Why animals move but plants stay put

Two reasons, which is why the answer is **"Both A and C"**:

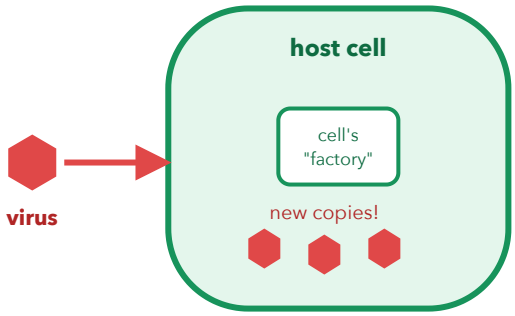
- **A:** Plants do not have muscles or legs to walk.
- **C:** Animals must go and **find** their food. Plants make their food right where they are.

REAL LIFE
A plant is like a kitchen that cooks its own meals. An animal has to go shopping!

4.3 Viruses: alive or not?

Viruses make us ill (a virus can even make a zebra ill!), but **most scientists say viruses are NOT living**.

- Viruses are **very small**, much smaller than a cell.
- Viruses are **not made of cells**. They are just a coat of **protein** with instructions (genetic material) inside.
- Viruses can only **replicate** (make copies of themselves) when they are **inside a living cell**.
- They **cannot carry out the seven life processes on their own**. They don't feed, grow, respire or excrete.



A virus invades a host cell and forces the cell to make copies of the virus.

THINK OF IT LIKE THIS
A virus is like a **pirate USB stick**. On its own it does nothing. Plug it into a computer (a cell), and it takes over the computer to make more copies of itself.

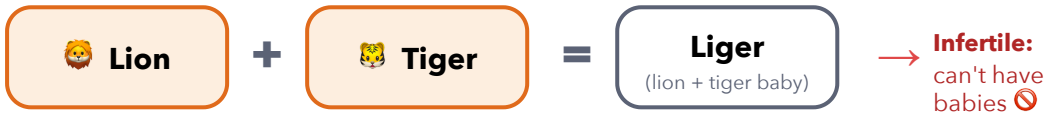
EXAM TIP

"Mia thinks a virus is a living organism. Is she correct?" Tick **No**. Explain: "A virus is not a living organism because it **cannot reproduce on its own**. It can only replicate inside a living cell, and it cannot carry out all seven life processes."

4.4 What is a species?

A **species** is a group of organisms that can **mate with each other** to produce **fertile offspring**. *Fertile* means the babies can grow up and have babies of their own.

- Organisms of the same species usually **look similar** to one another.
- They look **different** from organisms of other species.
- Every species has a two-part scientific name. The mountain zebra is *Equus zebra*.



EXAM TIP

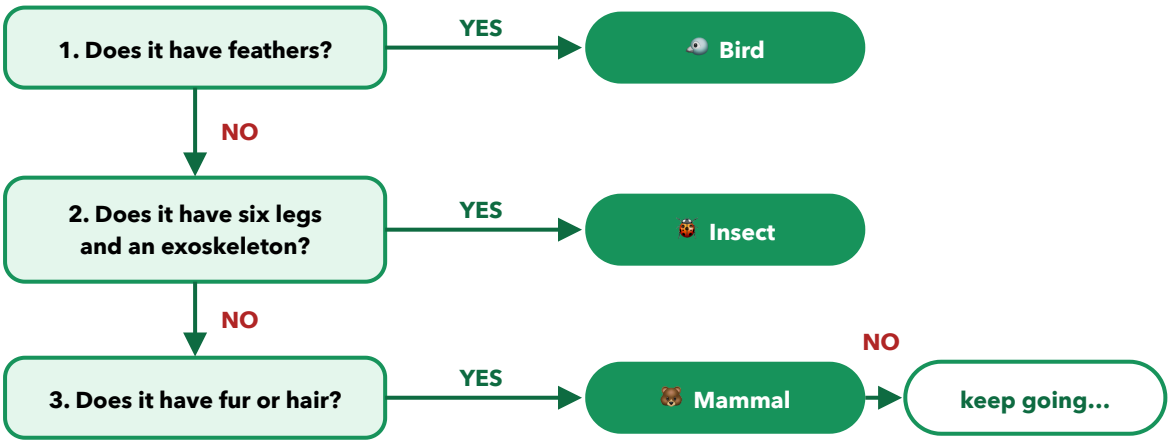
Why are lions and tigers different species, even though they can sometimes reproduce together?

"Organisms belong to the same species only if they reproduce to produce **fertile** offspring. Lions and tigers produce **infertile** offspring (ligers), which cannot reproduce." Use the word **fertile**!

4.5 Classification and dichotomous keys

Classification means putting living things into groups. Scientists do this to make it **easier to study and understand** the huge variety of life on Earth, a bit like a library sorting books into sections.

A **dichotomous key** helps you **identify an unknown organism**. At each step you choose between **two** options (a pair of features). The prefix **"di-"** means **two**, just like in *di*oxide (two oxygens).



A simple dichotomous key. Every question has exactly two answers.

- To separate **birds from mammals**, the best feature is **having feathers**. Both have eyes and mouths and need water, so those features don't help!
- An animal with **six legs and an exoskeleton** (a hard outer skeleton) is an **insect**.

EXAM TIP

A good key question uses a feature you can **see** and that splits the group into **two** clearly. "Is it cute?" is a bad question; "Does it have feathers?" is a good one.

4.6 The zebra question (Q7)

(a) Define species: a group of living things that can reproduce together to produce **fertile offspring**.
(b) Three other characteristics (besides growth) shown by the zebras: one is eating grass (**nutrition**), one is walking (**movement**), one has a foal (**reproduction**). The worksheet accepts **reproduction, respiration, movement**. Any three of the seven (except growth) are correct.
(c) Is a virus living? No. It cannot reproduce on its own.

✓ Unit 4 worksheet answers at a glance

Q	Answer	Why
1	B) Reproduction	Reproduction = making young ones.
2	A) Sunlight, water, carbon dioxide	The three ingredients of photosynthesis.
3	B) Can't do the 7 life processes on its own	Viruses need a host cell.
4	C) Invades a host cell, forces it to make copies	Viruses replicate only inside living cells.
5	C) Species	Same species = can make fertile offspring.
6	C) Identifying unknown organisms	You choose between pairs of features.
7	B) Two	"Di-" means two.
8	B) Nutrition	Taking in food for energy and growth.
9	B) Excretion	Getting rid of waste the body made.
10	B) Oak tree	Plants photosynthesise; a mushroom is a fungus.
11	D) Both A and C	No muscles/legs, and plants make food in place.
12	B) Having feathers	Only birds have feathers.
13	B) Insects	Six legs + exoskeleton = insect.
14	B) Permanent increase in size and mass	Growth must be permanent.
15	A) Easier to study the variety of life	Grouping makes life easier to understand.

Complete the sentences: nutrition = **feeding** · polar bears **sense** smells · animals excrete **carbon dioxide** · organisms **reproduce** · getting bigger = **growth** · breaking down food for energy = **respiration** · changing shape/position = **movement**.

Viruses word list: Viruses are very **small**. Viruses are not made of **cells**. Viruses can only **replicate** when they are inside a living cell.

Matching (Q6): nutrition → feeding · respiration → breaking down nutrients to release energy · excretion → getting rid of waste · reproduction → making new individuals · growth → getting bigger · sensitivity → detecting changes · movement → changing position.

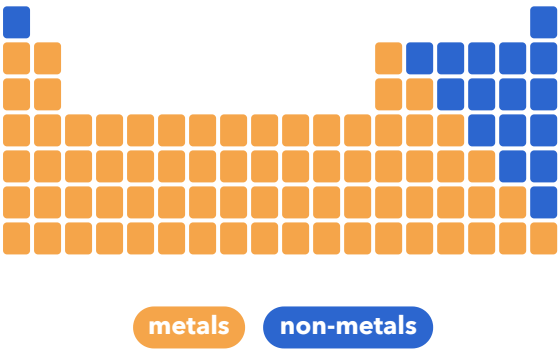
5 Properties of Materials

Why are pans made of metal but their handles made of plastic? Why is gold used for rings? What makes lemon juice sour?

Big idea: Every material has **properties** (what it is like and how it behaves). We choose a material for a job **because of its properties**. Most elements are **metals** or **non-metals**, and they have opposite properties.

5.1 Metals vs non-metals

On the **Periodic Table** (the big chart of all the elements), **metals** are on the **left and in the middle**. **Non-metals** are mostly in the **top right corner**.



Metals	Non-metals
Most are **solid** at room temperature	Many are **gases** at room temperature
Shiny	**Dull**
Conduct heat well (feel cold to touch)	Do **not** conduct heat well
Conduct electricity	Do **not** conduct electricity
Malleable & ductile	**Brittle** if solid
Sonorous (they ring)	Not sonorous
Strong and tough	Usually weak

THE TWO FAMOUS EXCEPTIONS

Mercury is a metal, but it is the **only metal that is liquid** at room temperature (it was used in old thermometers).
Carbon in the form of graphite (pencil "lead") is a non-metal that **can conduct electricity**.

EXAM TIP

"A material is **brittle, dull** and **does not conduct electricity**. Metal or non-metal?" → **Non-metal.** All three are non-metal properties.

Ten metals

Iron, gold, silver, copper, nickel, cobalt, mercury, aluminium, zinc, and the answer key also lists **steel**.

Five non-metals (other than sulfur & helium)

Carbon, oxygen, nitrogen, chlorine, phosphorus.

Uses of non-metals

- **Sulfur:** making **sulfuric acid** and **fertiliser**.
- **Helium:** balloons, because it is a **very light gas** and **does not react** easily, so it is safe (it won't catch fire).

GOOD TO KNOW

Steel is really an alloy (iron mixed with carbon), not a pure metal. Write what your teacher expects, but if you want to be extra safe, use **lead** or **tin** instead.

5.2 The property dictionary

These words come up again and again. Learn them and you can answer many questions!

Word	Meaning	Memory trick / example
Malleable	Can be **hammered** or rolled into flat **sheets**.	A **mallet** is a hammer → **mall**eable. Aluminium foil!
Ductile	Can be pulled or drawn into thin **wires**.	Think "**duct** tape on a **wire**". Copper wire.
Brittle	Breaks or **shatters** easily when hit.	Glass, sulfur, a biscuit.
Sonorous	Makes a **ringing sound** when hit.	**Son**orous sounds like **son**g. Brass bells.
Hardness	How well it resists being **scratched or dented**.	Diamond is the hardest natural material.
Strength	How much **pulling or pushing force** it can take before breaking.	Steel cables hold up bridges.
Density	How much **mass** is packed into a given **volume**.	A cube of lead feels much heavier than the same-size cube of wood.
Flexible	Bends without breaking.	A rubber band.
Transparent	Lets light pass **straight through clearly**.	Window glass, clear plastic bottles.
Conductor	Lets heat or electricity pass through easily.	Copper, silver, aluminium, iron.
Insulator	Does **not** let heat or electricity pass through easily.	Wood, plastic, rubber, air.
Magnetic	Attracted to a magnet.	**Only** iron, steel, nickel, cobalt. Gold, silver, copper and aluminium are **not** magnetic!

MEMORY TRICK

Magnetic metals: **"I Should Never Cry"**: **I**ron, **S**teel, **N**ickel, **C**obalt.

5.3 Choosing the right material for the job

Use	Material	Because it is...
Electrical wiring in homes	**Copper**	a good **electrical conductor** (and ductile)
Building bridges	**Iron** (steel)	**strong and tough**

Jewellery	**Gold, silver**	**shiny**, attractive and **do not corrode** easily
Aeroplanes	**Aluminium**	**lightweight**, **strong** and **resistant to corrosion**
Pot handles, tool handles	**Plastic**	a good **thermal and electrical insulator**
Cooking spoons	**Wood**	a good **thermal insulator** (won't burn your hand)
Balloons	**Helium**	very **light** and **unreactive**

WATCH OUT!

Silver is actually a better conductor than copper, so why don't we use it for wires? Because silver is **too expensive**!

CIRCLE THE METAL PROPERTIES

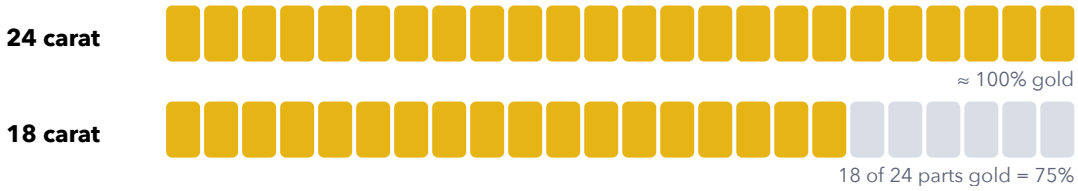
Metal properties: **malleable, ductile, feels cold to the touch, makes a ringing sound when tapped, conducts heat, has a shiny surface**. NOT metal properties: has a dull surface, does not conduct electricity, brittle.

5.4 Alloys and carat gold

An **alloy** is a **mixture of a metal with one or more other elements** (often other metals). Mixing makes the metal better: stronger, harder or less likely to rust.

- **Steel** = iron + carbon (stronger than iron)
- **Brass** = copper + zinc (sonorous, used for bells and trumpets)

Pure gold is very soft, so jewellery gold is mixed with other metals. Gold purity is measured in **carats**, out of **24**.



24-carat gold is nearly pure gold. 18-carat gold is 18 parts out of 24, which is **75% gold**.

HOW TO WORK IT OUT

18 ÷ 24 = 0.75 = **75%**. (And 12-carat gold would be 12 ÷ 24 = 50%.)

5.5 Changes to metals: rusting and expanding

Rusting (corrosion)

When **iron** is left in **air** (oxygen) and **moisture** (water) over time, it reacts and forms orange-brown **rust**. This is a chemical reaction called **rusting**, a type of **corrosion**.

iron + oxygen + water → rust

Expansion

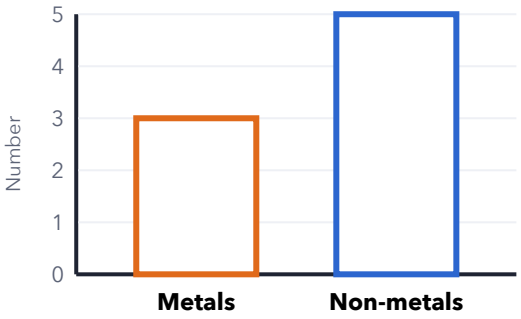
When most solid metals are **heated**, they **expand** (get slightly bigger). The particles **vibrate more** and push each other further apart.

WATCH OUT!
The particles do **not** get bigger; they just take up more space.

5.6 Sofia's metal mix-up

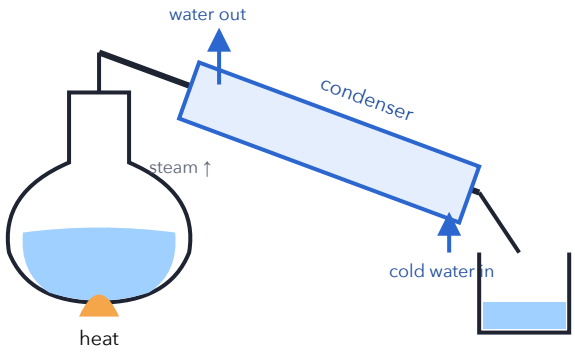
Material	Sofia said	Correct?
Silver	Metal	✅
Oxygen	Non-metal	✅
Helium	Non-metal	✅
Carbon	Metal	❌ (non-metal)
Copper	Metal	✅
Nitrogen	Non-metal	✅
Sulfur	Metal	❌ (non-metal)
Iron	Metal	✅

Sofia thinks there are **5** metals and **3** non-metals. Really there are **3 metals** (silver, copper, iron) and **5 non-metals**.



Bar chart tips: use a ruler, label both bars, number the axis, and **don't shade** the bars if the question says not to.

5.7 Boiling and condensing



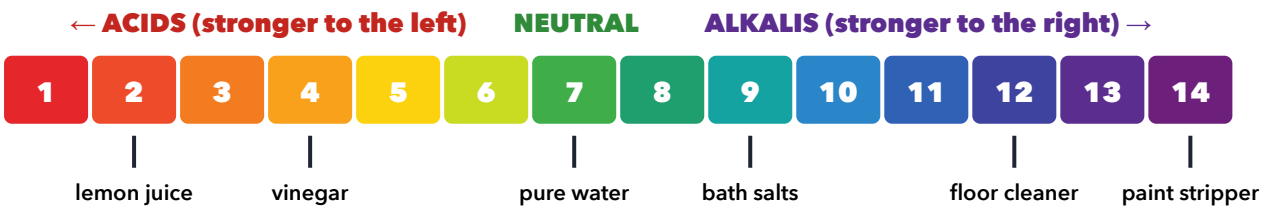
Boiling: the water in the flask **absorbs heat energy** and boils into **steam** (a gas) at **100 °C**.

Condensing: the steam travels into the **condenser**. **Cold water** flowing around the outside cools the steam, so it **condenses** back into **liquid water** and drips into the beaker.

REAL LIFE
Breathe on a cold mirror: your warm breath condenses into tiny water droplets.

5.8 Acids, alkalis and the pH scale

The **pH scale** goes from **1 to 14** and tells us how acidic or alkaline a liquid is.



- **pH 1–6 = acid.** The **lower** the number, the **stronger** the acid. pH 1 is **strongly acidic**.
- **pH 7 = neutral.** Pure **water** is neutral because its pH is **7**.
- **pH 8–14 = alkali.** The **higher** the number, the stronger the alkali.

Indicators

Universal indicator changes to a different colour for each pH: **red** for strong acids, **green** for neutral, **purple** for strong alkalis.

Litmus paper: blue litmus turns red in an acid. Red litmus turns blue in an alkali.

> **MEMORY TRICK**
> **A**cid turns litmus **red**. Think "**A**cid = **A**pple red".

Chen's pH chart answers

- pH of **vinegar** = **4**
- Paint stripper (pH 14) turns universal indicator **purple**
- pH of a **neutral** solution = **7**
- Most acidic = **lemon juice** (pH 2, the lowest number)

5.9 Hazard symbols and lab safety



Corrosive



Flammable



Hazardous to the environment

- **Corrosive** means a substance that can **damage skin tissues** and materials, causing **chemical burns**. Strong acids and alkalis are often corrosive.
- **Flammable** means it **catches fire easily**.
- **Hazardous to the environment** means it can harm plants, fish and nature.

> **EXAM TIP**
> **What should you do if you spill acid? → Wash the area with lots of water** (and tell your teacher).

> **DRAWING THE LABELS**
> All hazard symbols are a **red diamond** (a square on its point) with a **black picture** inside. Corrosive = liquid dripping from two test tubes onto a surface and a hand. Flammable = a flame.

✓ Unit 5 worksheet answers at a glance

Q	Answer	Q	Answer
1	B) Left and middle	10	A) Hardness
2	C) Mercury	11	B) Force it can take before breaking
3	B) Hammered into flat sheets	12	B) Mass packed into a given volume

4	A) Drawn into thin wires	13	A) Good electrical and thermal insulator
5	B) Brittle	14	B) Carbon (graphite)
6	B) Good electrical conductor	15	B) Makes a ringing sound
7	C) Wood	16	B) Rusting (corrosion)
8	B) Iron, steel, nickel, cobalt	17	B) Lets light pass straight through
9	B) Metal mixed with other elements	18	B) They expand as particles vibrate more

Short answers: Gold/silver jewellery: shiny, attractive, don't corrode · Copper: good conductor of heat and electricity · Helium: very light, unreactive · Aluminium planes: light, strong, corrosion-resistant · 18-carat = 75% gold, 24-carat ≈ pure · Water = neutral, pH 7 · pH 1 = strongly acidic · Blue litmus → red in acid · Spill acid → wash with lots of water.

9 Electricity

What is actually moving inside a wire? Why do lamps get dim? And why are plugs made of plastic AND metal?

Big idea: An **electric current** is a flow of tiny particles called **electrons**. Current only flows when there is a **complete loop** (a circuit) from the cell and back again.

9.1 What is electric current?

Electric current is the flow of **electrons** around a circuit. Electrons are tiny particles found in atoms. In **metals**, some electrons are **free to move**, which is why metals conduct electricity.

- The **greater the current**, the **faster the flow of electrons**.
- Current is measured in **amperes** (amps), with the unit **A**.
- Current is measured with an **ammeter**.

THINK OF IT LIKE THIS

A circuit is like a **bicycle chain**. The cell is the pedals, pushing the chain. The electrons are the links. When you pedal, **every link moves at once**, all the way round. Break the chain (open a switch) and everything stops.

WATCH OUT!

It's **electrons** that move, not atoms, protons or neutrons.

9.2 Components and circuit symbols

Scientists draw circuits using **symbols** instead of pictures. Learn these!

Symbol	Name	Its job in a circuit
	**Cell**	**Provides the energy** to make current flow. Long line = positive (+). Two or more cells = a **battery**.
	**Lamp**	Gives out **light**.
	**Switch (open)**	**Opens or closes** the circuit, to stop or allow the current. Open = OFF.
	Switch (closed)	The gap is closed, so current flows. Closed = ON.
	**Ammeter**	**Measures the current** in amps (A).

	**Buzzer**	Turns electrical energy into **sound** energy (makes a buzzing sound).

MEMORY TRICK

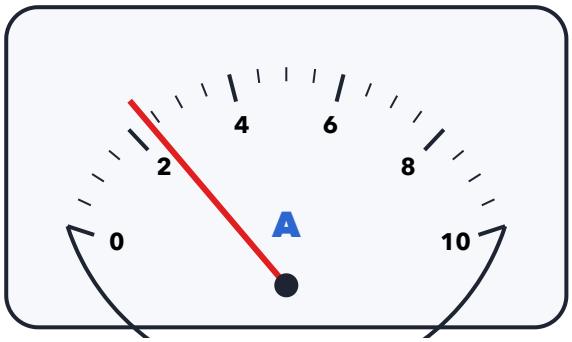
Ammeter measures **A**mps. Both start with A, and the symbol has an A inside it!

9.3 Measuring current

- The **ammeter** must be connected **in series**, meaning in the **same loop** as everything else, so all the current flows through it.
- A **multimeter** can measure different things. When Sofia switched hers to measure **current**, she read the **amps scale**.

READING A METER

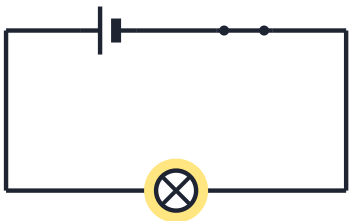
Find the scale for the thing you're measuring (here, the one marked **A**). See which two numbers the needle is between, then count the small marks. Sofia's needle was just past **2**, so the reading is **2.2 A**.



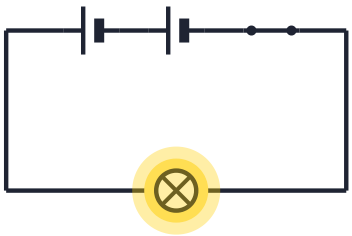
The needle is just past 2 → **2.2 A**

9.4 Series circuits

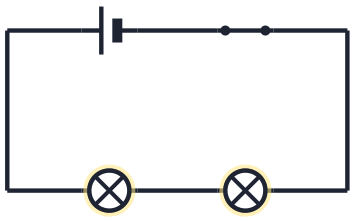
A **series circuit** has only **one loop**. Everything is connected one after another, like beads on a necklace.



1 cell, 1 lamp: normal



2 cells: BRIGHTER



2 lamps: DIMMER

If you...	What happens
Open the switch	The current **stops completely**. The loop is broken.
Add another cell	More push, so **more current**. Lamps shine **brighter**.
Add more lamps (same cells)	Harder for current to flow, so **less current**. All lamps get **dimmer**.
One lamp **breaks**	The loop is broken, so **all the other lamps go out**.
Measure current at different places	The current is **the same everywhere** in a series circuit.

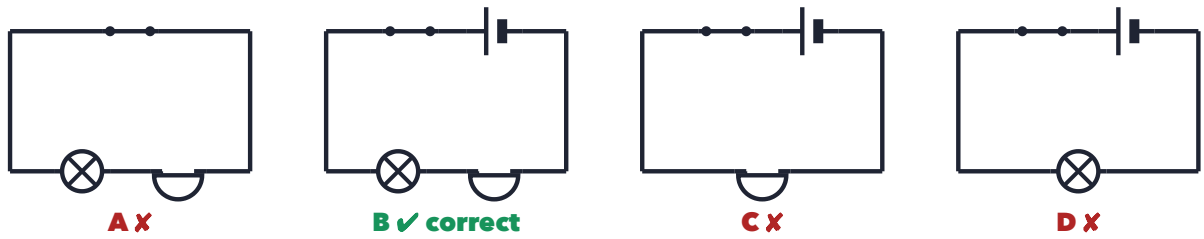
EXAM TIP

Circuit with two cells, a switch, three lamps and an ammeter:
To increase the current: (1) add more cells, (2) remove a lamp.
To decrease the current: (1) remove a cell, (2) add more lamps.

WATCH OUT!

Current is **not "used up"** as it goes round. It stays the same all the way around a series circuit. The energy is used, not the current.

Which circuit lights the lamp AND sounds the buzzer?



You need **three** things: a **cell** (energy), a **lamp** and a **buzzer**, all in one complete loop. Only circuit B has all three.

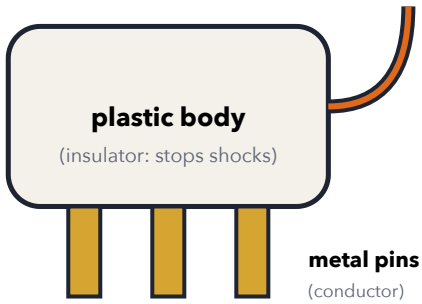
9.5 Conductors and insulators

A **conductor** lets electricity flow through it easily.
An **insulator** does **not**.

Conductors ⚡	Insulators 🛑
Copper, steel, aluminium, iron, silver, any **metal**, graphite	Plastic, wood, rubber, glass, **cotton**, air

IN TERMS OF PARTICLES

Conductors have **free electrons** that can move. **Insulators** do **not** have free electrons that can move easily.



A plug uses BOTH kinds of material.

- **Why are plug bodies made of plastic?** Plastic is an **insulator**, so it helps **prevent electric shocks**.
- **Why are plug pins made of metal?** Metal is a **conductor**, so electricity can flow into the appliance.
- **Why are wires coated in plastic?** Plastic is a good **insulator** that **prevents electric shocks**.
- **Which could NOT make electric wires? Cotton** (it's an insulator). Steel, copper and aluminium all conduct.

9.6 Lily's investigation

Lily's battery had two hazard symbols: **A = Corrosive** (it has chemicals that can burn skin) and **B = Hazardous to the environment** (dead tree and fish, so don't throw batteries in the bin!). The meter she used to measure current is an **ammeter**.

✓ Unit 9 worksheet answers at a glance

Q	Answer	Q	Answer
1	B) Electrons	9	B) All other lamps go out
2	B) Cell (or battery)	10	A) Conductor
3	B) Current stops completely	11	B) Copper
4	B) Amperes / amps (A)	12	B) Insulator
5	C) Ammeter	13	B) Plastic is an insulator, prevents shocks
6	A) In series	14	B) Switch
7	B) Shines brighter	15	B) Buzzer
8	B) All get dimmer		

Short answers: Particles that move = electrons · "The greater the current, the **faster** the flow of **electrons**" · "When current flows, particles called **electrons** move. In metals, these particles are **free to move**" · Silver not used for wires: too expensive · Current in series: stays the same · Good conductor: **metal** · Can't make wires: **cotton** · Symbols: cell, lamp, ammeter, (open) switch · Multimeter reading: **2.2 A**.

★ One-Page Cheat Sheet ★

Unit 4 · Living things

- **MRS GREN**: Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition
- Respiration = releasing energy from food (≠ breathing)
- Excretion = removing waste the body made (e.g. CO_2)
- Nutrition = feeding
- Growth = **permanent** increase in size & mass
- Photosynthesis needs **sunlight, water, CO_2**
- Mushroom = fungus, not a plant
- **Virus**: very small, not made of cells, replicates only inside a living cell → not living
- **Species**: can breed to make **fertile** offspring
- Lion + tiger = liger (infertile) → different species
- **Dichotomous key**: "di" = two; pairs of features
- Feathers → bird. 6 legs + exoskeleton → insect
- We classify to make life easier to study

Unit 5 · Materials

- Metals: left & middle of Periodic Table
- Metals: shiny, conduct heat & electricity, malleable, ductile, sonorous, strong
- Non-metals: dull, brittle, insulators, many are gases
- Mercury = liquid metal. Graphite = conducting non-metal
- **Malleable** = sheets · **Ductile** = wires · **Brittle** = shatters · **Sonorous** = rings
- Hardness = resists scratches. Strength = force before breaking. Density = mass per volume
- Magnetic: **I**ron, **S**teel, **N**ickel, **C**obalt
- Alloy = metal + other element(s). 18 ct = 75% gold
- Iron + air + water → rust. Heated metals expand
- pH 1–6 acid · 7 neutral (water) · 8–14 alkali
- Blue litmus → red in acid. UI: red = acid, green = neutral, purple = strong alkali
- Corrosive = burns skin. Spill acid → lots of water

Unit 9 · Electricity

- Current = flow of **electrons**
- Unit: **amps (A)**. Measured with an **ammeter**, connected **in series**
- Cell = energy · Switch = on/off · Lamp = light · Buzzer = sound
- Open switch → current stops
- More cells → more current, brighter
- More lamps → less current, dimmer
- One lamp breaks → all go out (series)
- Current is the **same everywhere** in series
- Conductors (metals) have **free electrons**; insulators don't
- Plug: plastic body (insulator), metal pins (conductor)
- Wires: copper, not silver (too expensive)
- Cotton can't make wires
- Lamp + buzzer circuit needs a cell in a complete loop

You've got this! Read the question twice, underline key words, and use science words in your answers. 🚀